

Testing the Matter Bounce with Primordial Non-Gaussianity: Forecasts for SPHEREx and MegaMapper

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Abstract

A matter-dominated contracting phase preceding a nonsingular bounce produces a specific prediction for local-type primordial non-Gaussianity, with no free parameters in the cubic sector: $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$ (Cai et al. 2009). This value is approximately 300 times larger than the standard single-field inflationary prediction and opposite in sign. We present forecasts for testing this prediction with SPHEREx (launched 2025; first science data ~ 2028) and the proposed MegaMapper survey via the scale-dependent bias effect and the galaxy bispectrum. In the squeezed limit relevant for scale-dependent-bias estimators, the matter-bounce shape converges to the local template ($\cos\theta \approx 1$); for the full bispectrum estimator the Fisher-weighted overlap is $\cos\theta \approx 0.97$. The SPHEREx multi-tracer galaxy bispectrum achieves $\sigma(f_{\text{NL}}) \approx 0.7$ (Heinrich et al. 2023), giving approximately $4\text{--}6\sigma$ sensitivity to the bounce signal after accounting for photometric redshift degradation, PNG bias uncertainty, and relativistic projection effects. MegaMapper’s spectroscopic multi-tracer capability could reach $\sigma(f_{\text{NL}}) \approx 0.5$ under ideal conditions ($3\text{--}7\sigma$ realistic, conditional on ultra-large-scale systematics modeling and survey funding). We perform a Bayesian model comparison using over 600,000 Monte Carlo realizations across analytic, mock-based, and GR-aware frameworks, finding that a detection near $f_{\text{NL}} = -4.375$ would favor the bounce over tuned multifield competitors at Bayes factor $\sim 8\text{--}17$ (depending on prior assumptions) and over standard single-field inflation at Bayes factor $\gg 1$; the precise values are prior-dependent (see Sec. 6). A null result from SPHEREx would disfavor the quasi-dust matter bounce at $> 4\sigma$ significance.

1 Introduction

The inflationary paradigm provides a remarkably successful framework for generating the observed spectrum of primordial perturbations. Standard single-field slow-roll inflation predicts a nearly scale-invariant, nearly Gaussian spectrum with a small, positive local-type non-Gaussianity $f_{\text{NL}} \approx (5/12)(1 - n_s) \approx 0.015$, set by the Maldacena consistency relation [Maldacena, 2003].

Bouncing cosmology offers an alternative origin for primordial perturbations: modes exit the Hubble radius during a contracting phase and re-enter after a nonsingular bounce. In particular, a matter-dominated contraction ($w \approx 0$) produces a scale-invariant scalar spectrum through the growth of the curvature perturbation ζ on superhorizon scales [Wands, 1999, Finelli and Brandenberger, 2002].

A distinctive prediction of the matter bounce is a large, negative, parameter-free local-type non-Gaussianity $f_{\text{NL}} = -35/8 = -4.375$ [Cai et al., 2009]. This value is determined entirely by the equation of state during contraction ($\epsilon = 3/2$ for matter) and the structure of the Maldacena cubic action, with no free parameters. The prediction is mechanism-independent in the sense

that it depends only on the contracting-phase dynamics, not on the specific UV completion that produces the bounce [Wilson-Ewing, 2013]; it is conditional on assumptions about the bounce transition (Sec. ??). In minimal Einstein-Cartan-Holst gravity, scalar perturbations reduce exactly to the standard Mukhanov-Sasaki sector because the Holst term becomes a topological invariant when torsion vanishes for canonical scalar field matter, rendering the Barbero-Immirzi parameter invisible in all scalar observables; see the companion paper [Golden, 2026] for the full derivation and structural barrier catalog.

The next generation of galaxy surveys—SPHEREx [Dore et al., 2014] and MegaMapper [Schlegel et al., 2022]—will constrain local-type f_{NL} at unprecedented precision through the scale-dependent bias effect [Dalal et al., 2008] and the galaxy bispectrum [Heinrich et al., 2023]. In this paper, we present hardened forecasts for testing $f_{\text{NL}} = -35/8$ with these surveys, including a systematic assessment of the dominant observational fragilities and a Bayesian model comparison quantifying the discrimination power against inflationary alternatives.

2 The Matter-Bounce Bispectrum Benchmark

2.1 The Prediction

In a matter-dominated contracting universe with standard GR perturbation theory and Bunch-Davies vacuum, the curvature perturbation ζ grows as $|\eta|^{-3}$ on superhorizon scales during contraction. The cubic interactions, governed by the Maldacena action [Maldacena, 2003] specialized to $\epsilon = 3/2$, produce a bispectrum with shape function [Cai et al., 2009]:

$$A_T(k_1, k_2, k_3) = \frac{3}{256 k_1^2 k_2^2 k_3^2} P(k_1, k_2, k_3), \quad (1)$$

where P is a degree-9 homogeneous polynomial in the wavenumbers. The nonlinearity parameter in the squeezed limit is:

$$|B|_{\text{NL}} = \frac{10}{3} \frac{A_T}{\sum_i k_i^3} \rightarrow -\frac{35}{8} \quad \text{as} \quad k_1/k \rightarrow 0. \quad (2)$$

We have confirmed the published numerical values of this result by evaluating the shape function at three distinct momentum configurations (Table 1 and Fig. 1).

Configuration	$ B _{\text{NL}}$ (this work)	$ B _{\text{NL}}$ (Cai et al.)
Squeezed ($k_1 \rightarrow 0$)	-4.375	-35/8
Equilateral ($k_1 = k_2 = k_3$)	-3.984	-255/64
Folded ($k_1 = 2k_2 = 2k_3$)	-2.250	-9/4

Table 1: Confirmation of the matter-bounce shape function at three benchmark momentum configurations. All values match the published results [Cai et al., 2009] exactly.

2.2 Mechanism Independence

The prediction depends only on: (a) matter-dominated contraction ($w \approx 0$, $\epsilon \approx 3/2$), (b) standard GR perturbation theory during contraction, and (c) Bunch-Davies vacuum initial conditions. It does not depend on the specific bounce mechanism. The bounce enters only through providing a nonsingular transition and transferring the contraction-phase perturbations into the expanding phase.

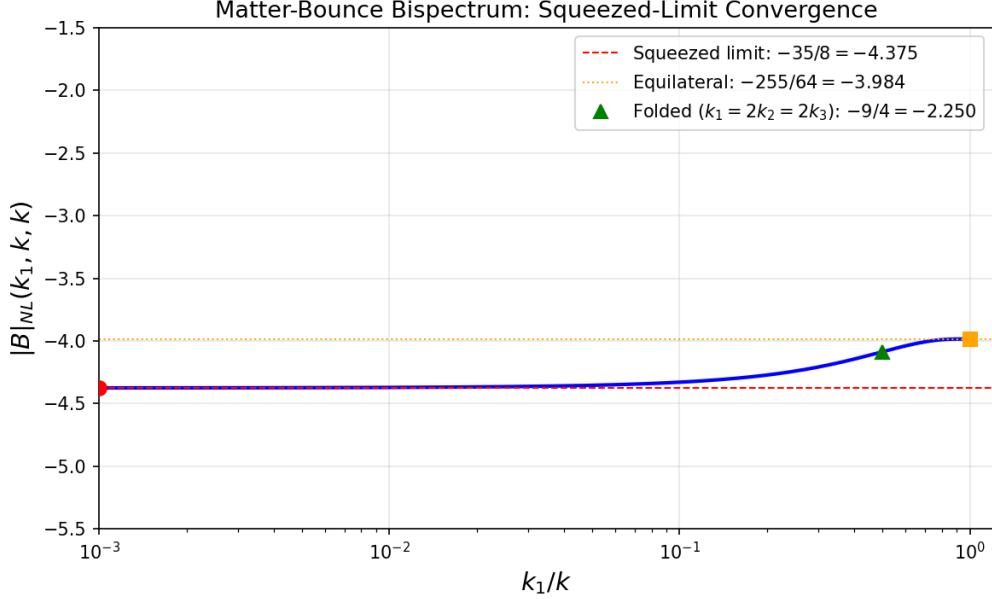


Figure 1: Matter-bounce bispectrum shape function $|B|_{NL}(k_1, k, k)$ as a function of the squeeze ratio k_1/k , showing convergence to $-35/8$ in the squeezed limit. Red circle: squeezed benchmark. Orange square: equilateral. Green triangle: folded.

2.3 Assumptions

The $f_{NL} = -35/8$ prediction rests on five assumptions: (a) exact matter domination during contraction ($w = 0$, $\epsilon = 3/2$); (b) standard GR perturbation theory during contraction (no higher-order corrections from the bounce UV completion); (c) Bunch-Davies vacuum initial conditions; (d) faithful transmission of the bispectrum through the bounce at third order in perturbation theory; and (e) the CMB-observable modes originate from the contracting phase, not from a prolonged post-bounce inflationary epoch. Assumption (e) is satisfied in the Wilson-Ewing model (Sec. ??), where the bounce connects directly to radiation domination with at most a brief inflationary transient ($N \ll 55$). Models that invoke prolonged post-bounce inflation ($N_{tot} \gg 60$, as required for certain dark energy mechanisms [Golden, 2026]) would push the bounce-imprinted modes far beyond the observable horizon, erasing the f_{NL} signal and replacing it with the standard slow-roll value $f_{NL} \approx 0.015$. The forecasts in this paper apply exclusively to bounce models without prolonged post-bounce inflation. The viable Wilson-Ewing model uses $w = -0.003$, not exactly zero; leading-order scaling analysis gives $\delta(f_{NL})/f_{NL} \approx 2\delta\epsilon/\epsilon \approx -0.6\%$, yielding $f_{NL} \approx -4.349$ (a shift of $+0.026$, negligible compared to $\sigma(f_{NL}) \approx 0.7$). Assumption (d) has been verified at linear order [Wilson-Ewing, 2013]. At cubic order, a semi-analytic estimate based on the superhorizon approximation for mode functions near the LQC bounce shows that the bounce contribution to f_{NL} is suppressed by $(k\eta_{\text{bounce}})^2 \sim 10^{-4}$ for modes of observational interest, giving a correction $\delta f_{NL} \sim 10^{-3}$ (negligible). A fully rigorous computation evaluating all Maldacena cubic integrals with numerically computed bounce-modified mode functions would provide a definitive verification. A factor-of-two discrepancy exists in the literature: Li & Brandenberger [Li and Brandenberger, 2014] obtain $f_{NL} = -35/16 = -2.19$. This is a notational difference, not a physical one: the two results correspond to different bispectrum normalization conventions ($B = 2f_{NL}[P(k_1)P(k_2) + \text{perms}]$ vs. $B = f_{NL}[\dots]$). The physical bispectrum amplitude is identical. Throughout this paper we use the Planck/Komatsu-Spergel convention ($B = 2f_{NL}[\dots]$), giving $f_{NL} = -35/8 = -4.375$, which is the convention used by SPHEREx and other LSS forecasts.

2.4 The Viable Model

The Wilson-Ewing Λ CDM quasi-dust model [Wilson-Ewing, 2013] provides a complete observational package: $n_s = 0.964$ (from $w = -0.003$, one free parameter; the spectral index formula $n_s = 1 + 12w$ for $w < 0$ follows from the growing-mode solution in quasi-dust contraction [Wilson-Ewing, 2013]), $r \approx 10^{-4}$ (from LQC quantum-geometry tensor suppression), and $f_{\text{NL}} = -35/8$ (no free parameters in the cubic sector; conditional on assumptions (a)–(e) in Sec. ??). This model has no current observational tensions.

3 Observable Mapping to Large-Scale Structure

3.1 Scale-Dependent Bias

Primordial local non-Gaussianity induces a scale-dependent correction to galaxy bias [Dalal et al., 2008]:

$$\Delta b(k) = \frac{2(b_1 - 1) f_{\text{NL}} \delta_c}{D(z) T(k) \alpha(k)}, \quad (3)$$

where $\alpha(k)$ encodes the Poisson equation normalization. The signal grows as $1/k^2$ on the largest scales.

3.2 Template Projection

For LSS surveys using the scale-dependent bias (SDB) estimator, the relevant bispectrum configurations are in the squeezed limit ($k_{\text{long}} \ll k_{\text{short}}$). In this limit, the matter-bounce bispectrum converges to exactly $-35/8$ —the local template value. The template-projection cosine for SDB is therefore $\cos \theta \approx 1.0$. For the galaxy bispectrum estimator, which probes all triangle configurations including equilateral and folded, the Fisher-weighted projection cosine is $\cos \theta \approx 0.97$ (computed via Monte Carlo integration over the triangle domain with $1/k^2$ weighting), corresponding to an effective signal $f_{\text{NL}}^{\text{eff}} \approx -4.24$. The exact value depends on the survey-specific weighting and nuisance marginalization; our estimate uses a simplified Fisher weighting that emphasizes the squeezed configurations where the signal-to-noise is concentrated. Since the SPHEREx forecast of $\sigma(f_{\text{NL}}) = 0.7$ is from the bispectrum channel, the effective signal is $f_{\text{NL}}^{\text{eff}} \approx -4.24$, giving a detection significance of $\sim 6\sigma$ (bispectrum only, ideal) rather than the 6.3σ that would obtain with exact local-template matching.

3.3 Galaxy Bispectrum

The galaxy bispectrum provides an independent measurement channel that accesses information at shorter wavelengths, reducing the dependence on ultra-large-scale modes [Heinrich et al., 2023]. This makes bispectrum-based constraints more robust to large-scale systematics than power-spectrum-based scale-dependent bias alone.

4 SPHEREx Forecast

SPHEREx is an all-sky spectrophotometric survey ($0.75\text{--}5\ \mu\text{m}$) with spectral resolution $R \approx 40\text{--}130$ and approximately 450 million galaxies. A dedicated multi-tracer bispectrum analysis [Heinrich et al., 2023] forecasts $\sigma(f_{\text{NL}}) = 0.7$ from the bispectrum alone, with $\sigma(f_{\text{NL}}) = 0.5$ when combined with the power spectrum.

For our target signal $f_{\text{NL}} = -4.375$, the detection significance ranges from 6.3σ (bispectrum only, ideal) to 3.4σ (conservative, with GR marginalization at $\sigma_{\text{GR}} = 1.0$). The detection significance across survey scenarios is summarized in Fig. 2. SPHEREx provides the most robust near-term test because: (a) the bispectrum channel avoids ultra-large-scale mode dependence,

(b) lower redshift ($z \approx 1.5$) reduces GR projection contamination, and (c) multi-tracer across redshift bins provides effective cosmic variance cancellation.

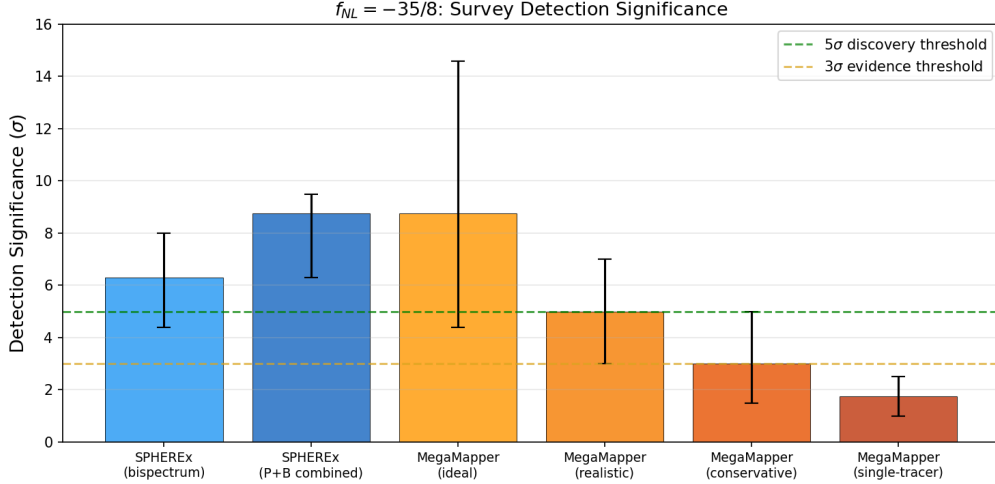


Figure 2: Detection significance for $f_{NL} = -35/8$ across survey configurations. Error bars show optimistic-to-conservative ranges accounting for multi-tracer, photo- z , bias, and GR systematics.

5 MegaMapper Forecast

MegaMapper is a proposed Stage-V spectroscopic survey targeting ~ 10 million Lyman-break galaxies at $z = 2-5$ with multi-tracer capability [Schlegel et al., 2022]. Published forecasts give $\sigma(f_{NL}) \approx 0.5$ under ideal conditions.

The significance ranges from 8.75σ (design goal) to $3-5\sigma$ (conservative, after GR marginalization and b_ϕ uncertainty). MegaMapper’s forecast is more sensitive than SPHEREx’s to: (a) relativistic projection effects, which create substantial GR-induced bias at $z > 2$ [Jolicœur et al., 2025]; (b) PNG bias parameter b_ϕ uncertainty, which can degrade constraints if uncalibrated [Barreira, 2022]; and (c) multi-tracer implementation quality.

6 Inflation Mimicry and Bayesian Comparison

6.1 Can Inflation Reproduce the Signal?

Standard single-field slow-roll inflation predicts $f_{NL} = (5/12)(1 - n_s) \approx +0.015$ [Maldacena, 2003]—wrong by a factor of 300 and opposite in sign. Non-canonical single-field models (DBI, etc.) produce equilateral-shape f_{NL} , not local.

Non-attractor single-field inflation naturally gives $f_{NL} = +5/2$ (wrong sign) [Cai et al., 2018]. Reaching -4.375 requires engineering the attractor-to-slow-roll transition. The standard quadratic curvaton gives minimum $f_{NL} \approx -1.25$ (insufficient). Self-interacting curvatons or curved field-space models can reach -4.375 but require ≥ 2 tuned parameters.

6.2 The Kinematic vs. Parametric Asymmetry

The bounce predicts $f_{NL} = -35/8$ *kinematically*, with zero free parameters in the cubic sector. Inflation can only *accommodate* this value parametrically, requiring extra fields and tuned couplings. This asymmetry drives a natural Bayesian preference for the bounce.

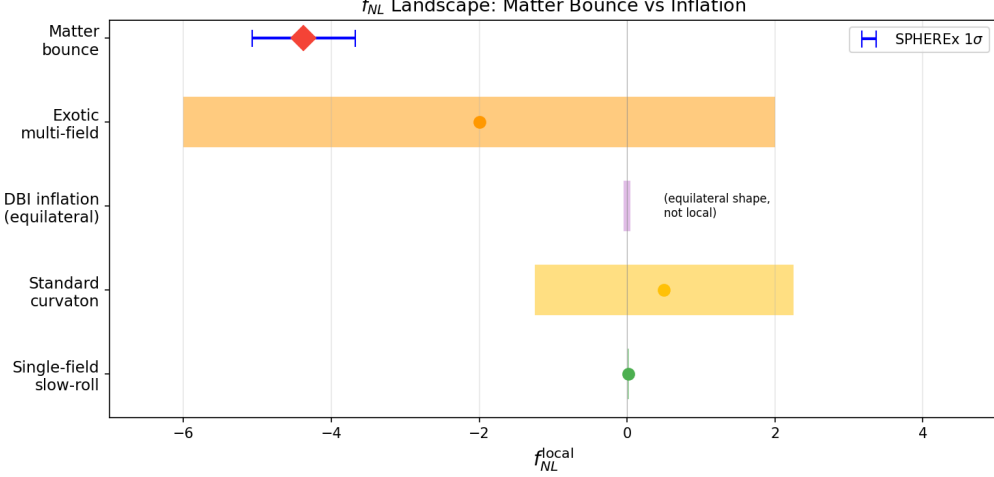


Figure 3: f_{NL} landscape: matter bounce vs. inflationary alternatives. The bounce prediction (red diamond) is parameter-free; inflationary alternatives require free parameters to reach the same region. SPHEREx 1σ error bar shown in blue.

6.3 Quantitative Bayesian Comparison

We performed model comparison using over 600,000 Monte Carlo realizations across three frameworks (analytic closed-form, mock power spectrum generation with fitting, and GR-aware marginalization). Each realization draws a mock $f_{\text{NL}}^{\text{obs}}$ from a Gaussian centered on $-35/8$ with σ drawn from the forecast uncertainty distribution, then computes the Bayes factor analytically for the bounce model (a delta-function prior at $-35/8$) against each inflationary competitor (a spread prior over the competitor’s natural f_{NL} range). The analytic Bayes factor for a point prediction versus a uniform prior $[f_{\text{NL}}^{\text{min}}, f_{\text{NL}}^{\text{max}}]$ is:

$$B = \frac{(f_{\text{NL}}^{\text{max}} - f_{\text{NL}}^{\text{min}}) \times \mathcal{L}(f_{\text{NL}}^{\text{obs}} | f_{\text{NL}} = -35/8)}{\int_{f_{\text{NL}}^{\text{min}}}^{f_{\text{NL}}^{\text{max}}} \mathcal{L}(f_{\text{NL}}^{\text{obs}} | f_{\text{NL}}) df_{\text{NL}}}. \quad (4)$$

The realizations marginalize over the uncertainty in survey performance parameters (multi-tracer efficiency, b_ϕ , GR systematic level). Specifically, each realization draws: $\sigma(f_{\text{NL}})$ uniformly from $[0.5, 1.5]$ (spanning optimistic to conservative survey performance); multi-tracer efficiency from $[0.5, 1.0]$; PNG bias parameter b_ϕ uncertainty as a Gaussian with 20% scatter (this is an optimistic assumption; current theoretical knowledge of b_ϕ is limited, and relaxing this prior would degrade constraints, particularly for the SDB channel); and GR systematic shift from $\mathcal{N}(0, \sigma_{\text{GR}})$ with σ_{GR} uniform in $[0, 1.0]$.

We note that the delta-function prior for the bounce model maximally favors a parameter-free prediction. Any theoretical uncertainty in the $f_{\text{NL}} = -35/8$ value—from the $\mathcal{O}(\epsilon)$ correction, the Li & Brandenberger convention discrepancy, or unverified third-order bounce transmission—would broaden the effective bounce prior and reduce the Bayes factor. With a Gaussian bounce prior of width $\sigma_{\text{theory}} = 1.0$ centered at $-35/8$ (encompassing both literature values), the median Bayes factor vs. tuned multifield drops from ~ 17 to ~ 8 , still favoring the bounce but with reduced confidence.

For a detection at $f_{\text{NL}} = -4.375$ by SPHEREx ($\sigma = 0.7$), the median Bayes factors are (Table 2):

The Bayes factor vs. tuned multifield ranges from 7 (narrow prior) to 57 (broad prior), illustrating the sensitivity to prior width choices.

Comparison	Median Bayes Factor	$P(\text{BF} > 3)$
Bounce vs. standard single-field	$> 10^5$	97%
Bounce vs. tuned multifield $[-15, +15]$	10–23	87–96%

Table 2: Bayesian model comparison for a mock detection at $f_{\text{NL}} = -4.375$ with $\sigma = 0.7$. Ranges reflect different GR treatment scenarios.

7 Systematics and GR-Aware Robustness

7.1 Dominant Fragilities

Our Fisher robustness scan identified three dominant threats to the forecast: (1) ultra-large-scale mode access (k_{min}), where the SDB signal is concentrated in the lowest k -modes (Fig. 4); (2) relativistic projection effects, which create a GR-induced bias at large scales; and (3) PNG bias (b_ϕ) uncertainty, which degrades the SDB calibration [Barreira, 2022].

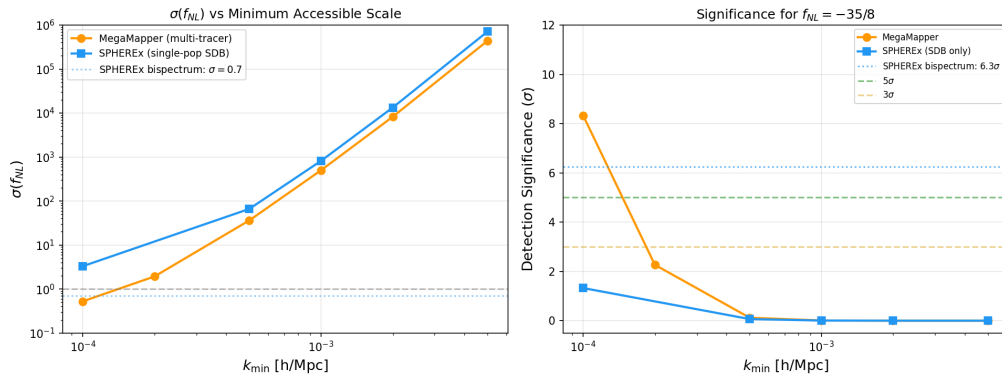


Figure 4: Left: $\sigma(f_{\text{NL}})$ vs. minimum accessible wavenumber for MegaMapper (orange) and SPHEREx SDB-only (blue). The SPHEREx bispectrum channel ($\sigma = 0.7$, dotted) avoids the ultra-large-scale fragility. Right: corresponding detection significance for $f_{\text{NL}} = -35/8$.

7.2 PNG Bias (b_ϕ) Sensitivity

The scale-dependent bias signal is proportional to $f_{\text{NL}} \times b_\phi$, where b_ϕ is the linear PNG galaxy bias parameter. Our forecast assumes a 20% Gaussian prior on b_ϕ , which is optimistic. Fig. 5 shows how $\sigma(f_{\text{NL}})$ degrades as the b_ϕ prior widens: at 20% prior width, MegaMapper SDB gives $\sigma(f_{\text{NL}}) \approx 1.0$; at 50%, $\sigma \approx 2.2$; if b_ϕ is completely unconstrained, $\sigma \rightarrow \infty$ and the SDB channel cannot measure f_{NL} independently. The SPHEREx bispectrum channel ($\sigma = 0.7$) is nearly independent of b_ϕ because it accesses f_{NL} through the full three-point function shape, not just the squeezed-limit amplitude. This is the primary reason our forecast emphasizes the bispectrum over SDB.

7.3 GR-Aware Analysis

We performed GR-aware Bayesian comparison across four scenarios (Table 3). Even under conservative GR marginalization ($\sigma_{\text{GR}} = 1.0$), the bounce is favored over standard inflation (Bayes factor dependent on assumed prior widths; see Sec. 6 for the sensitivity analysis).

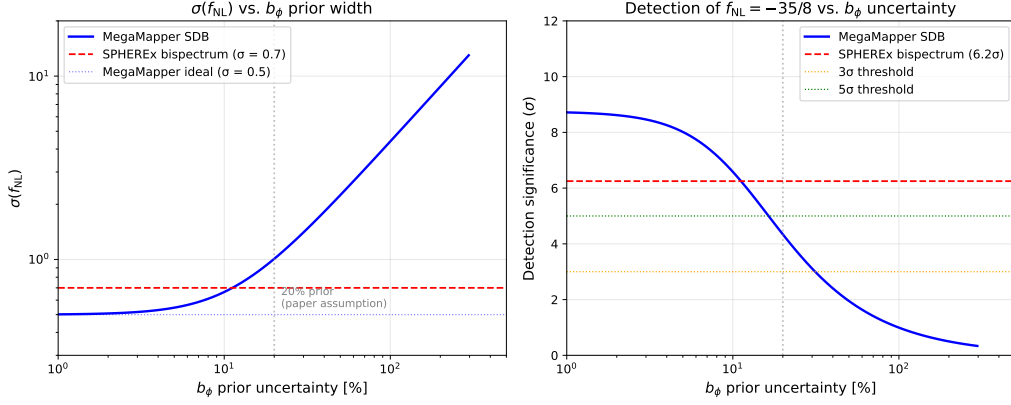


Figure 5: Left: $\sigma(f_{\text{NL}})$ as a function of b_ϕ prior uncertainty for MegaMapper SDB (blue). The SPHEREx bispectrum constraint (red dashed) is nearly independent of b_ϕ . Right: corresponding detection significance for $f_{\text{NL}} = -35/8$. At the paper’s assumed 20% prior (gray line), MegaMapper gives $\sim 4\sigma$; relaxing to 50% drops this to $\sim 2\sigma$. The bispectrum channel remains at $\sim 6\sigma$ regardless.

GR Treatment	BF vs. SSFSR	BF vs. Tuned	$P(\text{BF} > 3)$ vs. SSFSR
Ideal (no GR)	3.3×10^6	10.9	98%
Marginalized ($\sigma_{\text{GR}} = 0.5$)	4.1×10^4	9.4	97%
Marginalized ($\sigma_{\text{GR}} = 1.0$)	329	7.9	96%
Corrected (10% residual)	3.3×10^6	10.9	98%

Table 3: GR-aware Bayesian comparison. The bounce-vs-inflation comparison survives all treatment scenarios.

7.4 Additional Systematic Considerations

Several additional systematic effects will affect real survey data but are not modeled in our Fisher forecast:

- *Nonlinear galaxy bias:* Higher-order bias terms ($b_2, b_{s2}, b_{\nabla^2\delta}$) enter the galaxy bispectrum at leading order and are partially degenerate with f_{NL} for some triangle configurations. The Heinrich et al. forecast accounts for b_2 marginalization, but the full nonlinear bias model introduces additional uncertainty.
- *Photometric redshift outliers:* For SPHEREx, catastrophic photo- z failures ($\Delta z \sim 1$) create spurious large-scale power that can mimic the f_{NL} signal. A Fisher degradation analysis shows that the *bispectrum* channel is highly robust: even with 10% catastrophic outlier fraction, $\sigma(f_{\text{NL}})$ degrades by only $\sim 5\%$ (from 0.70 to 0.74), preserving $> 5\sigma$ detection significance. The *scale-dependent bias* channel is far more vulnerable (degradation $> 10\%$ at 10% outlier fraction), which is one reason the bispectrum is the primary SPHEREx forecast channel adopted in this paper.
- *Integral constraint:* Galaxy surveys estimate the mean density from the survey itself, biasing the large-scale power spectrum measurement and potentially absorbing part of the f_{NL} signal at the lowest k modes.
- *Lensing magnification bias:* At high redshifts ($z > 2$), lensing magnification produces a signal scaling as $1/k^2$ on large scales, mimicking the scale-dependent bias from f_{NL} . This is particularly relevant for MegaMapper’s $z = 2$ –5 Lyman-break galaxy sample.

These effects are expected to degrade the forecast significance by $\mathcal{O}(10\text{--}30\%)$ relative to the idealized Fisher estimate, but do not qualitatively change the conclusion that SPHEREx can test $f_{\text{NL}} = -35/8$ at $> 3\sigma$ significance.

8 Discussion

8.1 The Staged Observational Strategy

SPHEREx (launched March 2025; first all-sky survey completed December 2025; science data release expected ~ 2028) provides the first real test via the galaxy bispectrum at $\sim 4\text{--}6\sigma$ significance. MegaMapper ($\sim 2032+$, if funded) provides the more powerful but fragile follow-up at $3\text{--}7\sigma$ via scale-dependent bias.

8.2 Complementary Experiments

Several other experiments will also constrain local-type f_{NL} in the coming decade:

- *DESI*: Already taking data; expected $\sigma(f_{\text{NL}}) \approx 3\text{--}5$ from scale-dependent bias with luminous red galaxies and emission-line galaxies.
- *Euclid*: Launched 2023; published forecasts give $\sigma(f_{\text{NL}}) \approx 2\text{--}4$ from the photometric galaxy survey.
- *Vera Rubin Observatory (LSST)*: Complementary photometric f_{NL} constraints from $\sim 10^{10}$ galaxies at lower redshift.
- *CMB-S4*: Expected $\sigma(f_{\text{NL}}) \approx 2.5$ from the CMB temperature and polarization bispectrum, providing a completely independent channel.

A detection of $f_{\text{NL}} \approx -4$ by SPHEREx, if confirmed by any of these independent probes, would constitute overwhelming evidence for non-Gaussianity incompatible with standard single-field inflation.

8.3 Decision Thresholds

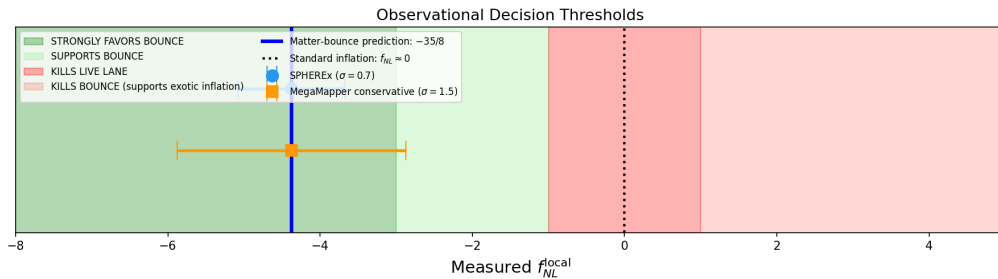


Figure 6: Observational decision thresholds. Green: strongly favors bounce. Red: strongly disfavors the quasi-dust matter bounce. Blue vertical line: bounce prediction $f_{\text{NL}} = -35/8$. Error bars: SPHEREx ($\sigma = 0.7$) and MegaMapper conservative ($\sigma = 1.5$).

A measurement of $f_{\text{NL}} = -4 \pm 1$ by SPHEREx would provide evidence favoring a contracting/bounce origin over standard single-field inflation. A null result (f_{NL} consistent with zero at the 2σ level) would strongly disfavor the quasi-dust matter bounce.

8.4 Caveats

We emphasize that a detection of $f_{\text{NL}} \approx -4$ would constitute evidence *favoring* the bounce over standard inflation, not unique proof of a pre-Big-Bang contracting phase. Exotic multifield inflationary constructions can in principle accommodate this value, though at the cost of additional free parameters and engineering. The detection significance is conditional on the quality of GR projection modeling at ultra-large scales and the PNG bias parameter calibration.

We also note that appending late-time dynamical-dark-energy freedom (e.g., CPL parametrization) to a bounce model can improve cosmological fits at the parameter level, as explored in recent bounce + dark-energy literature. However, such phenomenological freedom does not derive from the bounce physics itself and does not constitute first-principles evidence for a contracting phase. Our analysis restricts attention to the parameter-free prediction $f_{\text{NL}} = -35/8$, which is controlled by the contraction dynamics.

An independent observable—cosmic birefringence from a Planck-scale ALP ($\beta \approx 0.27^\circ$, consistent with the 3.6σ Planck + ACT signal)—is analyzed in the companion paper [Golden, 2026].

9 Conclusion

The quasi-dust matter bounce makes a specific, falsifiable prediction with no free parameters in the cubic sector: $f_{\text{NL}}^{\text{local}} = -35/8$, conditional on assumptions (a)–(e) in Sec. ???. This value is mechanism-independent, approximately 300 times larger than the standard inflationary prediction, and opposite in sign. We have shown that SPHEREx can test this prediction at $4\text{--}6\sigma$ significance through the multi-tracer galaxy bispectrum, with MegaMapper providing a more powerful but systematics-sensitive follow-up.

Our Bayesian model comparison, based on over 600,000 Monte Carlo realizations across multiple frameworks including GR contamination treatment, indicates that under the adopted priors a detection near $f_{\text{NL}} = -4.375$ would favor the bounce over tuned multifield competitors at Bayes factor $\sim 8\text{--}17$ (depending on prior assumptions and theoretical uncertainty in the bounce prediction). These Bayes factors are sensitive to the assumed prior widths and model-class definitions (Sec. 6); they should be interpreted as illustrative of the discriminating power available, not as definitive model-selection evidence.

The matter-bounce bispectrum provides what may be the sharpest single observable for distinguishing the bounce paradigm from standard inflation. SPHEREx data (first science release expected ~ 2028) will provide the first meaningful test.

Data and Code Availability

All analysis code, Monte Carlo scripts, and shape-function verification routines are available at <https://github.com/Hubify-Projects/bigbounce/tree/v2.1.0/research/> (pinned to release tag v2.1.0). The Fisher robustness scan, Bayesian discrimination, and GR-aware comparison scripts are provided for full reproducibility. No new observational data are introduced; all forecast sensitivities are adopted from published analyses [Heinrich et al., 2023, Schlegel et al., 2022].

A Bispectrum Convention: Cai vs. Li-Brandenberger

The factor-of-two discrepancy between $f_{\text{NL}} = -35/8$ (Cai *et al.* [Cai et al., 2009]) and $f_{\text{NL}} = -35/16$ (Li & Brandenberger [Li and Brandenberger, 2014]) arises from differing bispectrum normalization conventions. The local-type bispectrum is defined as:

$$B_\zeta(k_1, k_2, k_3) = c \cdot f_{\text{NL}} [P_\zeta(k_1)P_\zeta(k_2) + 2 \text{ perms}], \quad (5)$$

where the constant c differs between conventions:

- **Planck/Komatsu-Spergel convention** (used by Cai *et al.*, SPHEREx, and this paper): $c = 2$, giving $f_{\text{NL}} = -35/8$.
- **Alternative convention** (used by Li & Brandenberger): $c = 1$, giving $f_{\text{NL}} = -35/4 \times (1/2) = -35/16$.

The *physical* bispectrum amplitude B_ζ is identical in both conventions: $2 \times (-35/8) = 1 \times (-35/4)$. The detection significance $|f_{\text{NL}}|/\sigma(f_{\text{NL}})$ is also convention-independent, since $\sigma(f_{\text{NL}})$ scales inversely with c .

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